

MedTrack – Milestone 2

Member 6: Advanced Tableau Dashboard Storyboard & KPI Specification

Storyboard Architecture: Purpose → Standard & Novel KPIs → Advanced Tableau Visuals → Slicers & Parameters → Tableau Actions

Dashboard 1: Hospital Overview & Demand Intake Horizon

Purpose: Provide hospital executive leadership with real-time operational situational awareness, measuring inpatient admission throughput, bed saturation, and critically, unserved community healthcare demand.

Key Performance Indicators (KPIs)

Total Admissions (1,000) • Average Length of Stay (7.41d) • Bed Occupancy Rate (20.47%) • Bed Utilization Rate (92.47%) • **[Novel KPI 1] Patient Turnaway Rate (56.64%) • [Novel KPI 2] Demand Fulfillment Rate (43.36%)**

Tableau Visualizations (Advanced Differentiated Design)

Visual Name	Tableau Chart Type	What It Shows & Why It Differs from Standard Storyboard
Demand vs Capacity Horizon	Dual-Axis Area & Horizon Ribbon	Replaces simple line chart. Plots SUM(patients_request) vs SUM(patients_admitted), dynamically shading the 56.6% Unmet Turnaway Gap in red.
Department Intake Volume	Hierarchical Treemap w/ Sparklines	Replaces basic horizontal bar. Volume sized by tile area, with Viz-in-Tooltip sparkline revealing 12-week intake trajectory.
Hospital Bed Allocation	100-Mark Waffle Matrix & Radial Gauge	Replaces low-density donut chart. 10×10 grid of discrete marks displaying active occupancy (Emergency 100%, Surgery 88%) with instant status encoding.
Intake Channel Flow	Diverging Stacked Intake Stream	Replaces generic 2-bar chart. Displays proportion of Emergency (unscheduled) vs Elective cases across departments.
ALOS Target Benchmark	Clinical Target Bullet Graph	Replaces plain bar chart. Compares department average stay directly against the CMS/National clinical target benchmark (7.0 days).

Tableau Filters & Parameters

Year Filter • Admission Date Range Slider • Department Dropdown • Ward Type (ICU/General/Emergency) • Admission Type • Disease Category

Tableau Actions & Dynamic Interactions

- **Filter Action:** Clicking a department in the Treemap cross-filters the Demand Horizon and Bed Waffle.
- **Parameter Action:** Toggling date period dynamically updates the rolling moving average ribbon.
- **Viz in Tooltip:** Hovering over the Waffle Matrix displays ward-level bed turnaround times and nurse staffing.

Dashboard 2: Patient Flow, Clinical Acuity & Care Quality

Purpose: Analyze inpatient clinical trajectories, diagnosis distributions, care experience, and readmission patterns to identify clinical bottlenecks and improve care outcomes.

Key Performance Indicators (KPIs)

Total Inpatients (1,000) • Average Length of Stay (7.41d) • 30-Day Readmission Rate (0.00%) • **[Novel KPI 3] Net Patient Satisfaction Score (80.0/100)** • **[Novel KPI 4] Caregiver Morale vs Stress Index (72.57/100)**

Tableau Visualizations (Advanced Differentiated Design)

Visual Name	Tableau Chart Type	What It Shows & Why It Differs from Standard Storyboard
Clinical Progression Pathway	Patient Acuity Sankey / Diverging Flow	Replaces repeated horizontal bar. Maps flow from Intake Source → Department → Disease Diagnosis → Discharge Disposition.
Length of Stay Variance	Box-and-Whisker Distribution Plot	Replaces repeated plain bar. Shows median stay, IQR, and flags long-stay outliers (>14 days) per specialty instead of hiding variance in an average.
Readmission Surveillance	Statistical Process Control (SPC) Funnel	Replaces plain bar chart. Plots readmission rates against patient volume with 95% & 99.7% confidence limits to differentiate systemic vs special-cause variation.
Disease Acuity Matrix	Disease Severity Bubble Matrix	Replaces plain bar chart. X = Patient Volume, Y = ALOS, Size = Patient Satisfaction, Color = Disease Category, highlighting high-cost clusters.
Demographic Severity	Population Pyramid & Heat Strip	Replaces plain donut chart. Compares age/gender demographics crossed with admission urgency (Emergency vs Elective).

Tableau Filters & Parameters

Year Filter • Disease Category Slicer • Clinical Specialty • Age Bracket Slicer • Discharge Status Filter

Tableau Actions & Dynamic Interactions

- **Set Action:** Selecting a disease category isolates associated clinical stay profiles across all views.
- **Outlier Drilldown:** Clicking on an outlier in the Box-Whisker triggers a detailed patient case sheet.
- **Hover Highlight:** Hovering on any patient segment highlights their corresponding demographic bracket.

Dashboard 3: Bed & Resource Intelligence (4-Quadrant Strategic Efficiency)

Purpose: Monitor physical bed capacity, clinical workforce saturation, and strategic resource allocation across hospital departments.

Key Performance Indicators (KPIs)

Total Operational Beds (415) • Bed Occupancy (20.47%) • Bed Utilization (92.47%) • **[Novel KPI 5] Clinical Workforce Stress Index (88.98%)** • Strategic Department Efficiency Score (88.45/100)

Tableau Visualizations (Advanced Differentiated Design)

Visual Name	Tableau Chart Type	What It Shows & Why It Differs from Standard Storyboard
Strategic Efficiency Matrix	4-Quadrant Strategic Matrix (Scatter)	Replaces 1D ranked bar. Plots Bed Utilization % (X) vs Demand Fulfillment % (Y), with bubble size = Workforce Stress Index. Classifies depts into Overburdened vs Optimal.
Capacity Thresholds	Capacity Utilization Bullet Chart	Replaces generic horizontal bar. Displays actual bed utilization against calibrated safety bands: Safe (<80%), Optimal (80-85%), High Risk (85-95%), and Critical Overflow (100%).
Ward Strain Temporal Grid	Ward Strain Heatmap Calendar	Replaces static bed count bar. 2D heatmap matrix (Day of Week × Shift) colored from calm teal to urgent coral highlighting peak surge windows.
Bed Turnover Velocity	Turnaround Interval Velocity Gauge	Replaces stacked bar chart. Tracks idle bed turnaround hours between discharge and next admission.
Frontier Iso-Capacity Plot	Dynamic Iso-Capacity Scatter Curve	Replaces standard scatter plot. Adds curved iso-utilization boundary lines highlighting surplus-bed units vs severely constrained units.

Tableau Filters & Parameters

Resource Category Selector (Beds, Doctors, Nurses) • Department Selector • Shift Parameter (Morning/Afternoon/Night)

Tableau Actions & Dynamic Interactions

- **Quadrant Selection Action:** Clicking the 'Overburdened' quadrant filters shift-level staffing and bed logs.
- **Parameter Action:** Toggling the Resource Category parameter instantly updates bullet charts from Bed to Clinical Staff.

Tableau Story Navigation & Relational Data Integration

Navigation Flow: Dashboard 1 (Hospital Overview) → Dashboard 2 (Clinical Flow) → Dashboard 3 (Strategic Resource Matrix), connected via Tableau Navigation Buttons and Story Tabs.

Dashboard Page	Executive Strategic Question Answered in Tableau
Dashboard 1: Hospital Overview	What is the hospital's overall throughput, and where are we turning patients away due to capacity shortfalls?
Dashboard 2: Patient & Clinical Flow	How do clinical trajectories, disease severity, and stay duration impact patient care quality and caregiver morale?
Dashboard 3: Bed & Resource Matrix	Which departments operate in an overburdened vs underutilized quadrant, and how is workforce stress distributed?

Relational Dataset Fields Connected in Tableau Data Model

- **Hospital Overview Dataset:** admission_id, patient_id, name, age, department_name, admission_date, discharge_date, length_of_stay_days, patient_satisfaction_score, year, month, quarter, day_of_week, readmission_flag
- **Department Analytics Dataset:** year, week, department_name, patients_admitted_count, avg_length_of_stay_days, avg_satisfaction_score, patients_discharged_count, readmission_count, readmission_rate_pct, available_beds_ref, bed_occupancy_rate_ref_pct
- **Weekly Services & Demand Dataset:** week, month, service, available_beds, patients_request, patients_admitted, patients_refused, patient_satisfaction, staff_morale, event
- **Resource Utilization Dataset:** resource_utilization_id, week, department_name, resource_type, resource_category, total_units_available, units_in_use, utilization_rate_pct

Tableau Implementation Notes & Competitive Differentiation:

This deliverable replaces the repetitive single-bar and donut visuals in the initial Member 6 storyboard with publication-quality Tableau analytical patterns: dual-axis horizon area ribbons, 100-cell waffle matrices, clinical target bullet graphs, and a 4-quadrant strategic matrix. It integrates 5 novel performance indicators (Turnaway Rate, Demand Fulfillment Rate, Clinical Workforce Stress Index, Staff Morale, and Net Care Experience) directly into the Tableau workbook data model.